PROJECT DESIGN PHASE

Data Flow Diagrams and User Stories

Streamlining IT procurementbyautomatingstandardlaptoporderswithFlowDesigner

Session detail	Working value	Session detail	Workingvalue
Date	1 October 2026	Team / owner	SWTID-2026-2670
Project	Standard laptop order automation	Scope	Routine, policy-compliant requests

How to use this template

Start with the context diagram, then decompose the workflow into processes and data stores. Use the story examples as candidate backlog items and confirm them with requesters, IT, procurement, finance, security, and fulfillment owners. The source document's diagrams and mobile-app examples are illustrative instructions, not facts about this project.

DFD Level 0 Context view

Conceptual boundariesonly. Replace or confirmeach external system, data direction,field, source of truth, and owner during discovery. The centralprocess represents the proposed FlowDesigner orchestration; connectedsystems are not yet confirmed.

External entity / system	Data sent to process	Central process	Data returned or sent onward	External entity / system
Employee / manager	Laptop choice, business need, delivery details, requester identifiers ->	STANDARD LAPTOP ORDER WORKFLOW Flow Designer orchestration (platform instance and interfaces to confirm) Eligibility and completeness checks; policy-based routing	Confirmation, status, correction request <-	Employee / manager
Worker or identity source [TBD]	Worker, role, location, cost context ->	Request, validation, approval and fulfillment coordination Workflow status and exception handling	Approval task / decision <->	Authorized approver
Approved catalog / inventory [TBD]	Standard models, availability, lead time <->		Approved purchase request ->	Procurement / purchasing system [TBD]
Procurement / purchasing [TBD]	Order, supplier, PO or status <->		Receipt, serial, asset assignment <->	IT fulfillment / asset source [TBD]

Boundary and notation notes

Clarify	Record
Included process start and end	[e.g. request submitted through device receipt and asset handoff; confirm]
Arrow direction	A -> B means data moves from A to B; <-> means exchange in both directions. Confirm actual interfaces.
Out of scope / separate flows	[Non-standard orders, urgent replacements, exceptions, or other workflows]

DFD Level 1 Process and data-store worksheet

Break the context process intonumbered transformations. Name onlyconfirmed data stores and integrations; use TBD where ownership or system of record is unknown.

ID / process	Inputs	Process / rule	Output	Data store or system	Exception / failure path
1. Capture request	[Requester, model, business need, location, need date]	[Required fields and submission checks]	[Request record and confirmation]	[Request platform / store TBD]	[Missing, duplicate, or conflicting request]
2. Validate eligibility	[Request plus worker, role, location, assignment data]	[Approved catalog and eligibility rules]	[Eligible result or exception reason]	[Directory / HR, asset source TBD]	[Data unavailable or eligibility uncertain]
3. Route approvals	[Validated request and policy context]	[Policy-defined approval routing]	[Decision, approver, timestamp]	[Approval record / audit store TBD]	[Reject, timeout, delegation, escalation]
4. Source / purchase	[Approved configuration, cost, delivery details]	[Check supply; create or hand off purchase request]	[Order reference, expected timing]	[Catalog / inventory / procurement TBD]	[Out of stock, integration failure, budget hold]
5. Fulfill and track	[Order and shipment / receipt updates]	[Coordinate receipt, configuration, delivery]	[Status milestones and delivery confirmation]	[Supplier / fulfillment source TBD]	[Delay, lost shipment, changed delivery]
6. Assign and close	[Serial, asset, assignee, handoff confirmation]	[Record asset and close request]	[Assignment and closure notice]	[Asset management source TBD]	[Missing serial, setup failure, unconfirmed handoff]

Data-store and ownership register

Data set / store	System of record	Data owner	Read / write	Classification / retention notes
Worker identity, role, location	[TBD]	[TBD]	[R/W/TBD]	[Confirm minimum necessary fields]
Catalog, eligibility, policy	[TBD]	[TBD]	[R/W/TBD]	[Version and policy owner]
Request, approval, audit trail	[TBD]	[TBD]	[R/W/TBD]	[Access, retention, audit]
Purchase, inventory, asset	[TBD]	[TBD]	[R/W/TBD]	[Supplier, serial, assignment controls]

DFD review checks

-Every external entity, process, data store, and flow has a clear name; each arrow has a direction and meaningful data label.
-Data crossing the system boundary is necessary for the step; confirm access, retention, audit, and failure handling with owners.
-Show rejected requests, manual exceptions, retries, and disconnected-system paths, not only the happy path.

Candidate user stories

These arestarter hypotheses for discussion, notapproved requirements. Replace TBD rules with policy-approved conditions. “Priority” and “Release” remain for the product andprocess owners to decide.

User type	Epic / requirement	ID	User story and acceptance criteria	Priority	Release / dependency
Employee / manager	Guided request	US-01	As an employee or manager, I want to select an approved standard laptop option and provide the required request details, so that I can submit a complete request. Given the published standard catalog and required fields, when I submit a valid request, then a request record is created and I receive its reference. When required information is missing, then I am told what to complete before submission.	TBD	TBD / catalog, form
Employee / manager	Request visibility	US-02	As a requester, I want to see the current request stage and next expected action, so that I know whether I need to do anything. Given a submitted request, when its state or owner changes, then the requester can see the approved status and receives agreed milestone communications.	TBD	TBD / status source
IT / procurement	Eligibility validation	US-03	As a process owner, I want each request checked against agreed eligibility and catalog rules, so that routine requests are handled consistently. Given approved rule definitions and available source data, when the workflow validates a request, then it records the rule outcome; when data is missing or ambiguous, then it routes the request to a named exception path.	TBD	TBD / policy, worker and asset data
Approver	Policy-based approval	US-04	As an authorized approver, I want a review task with the request context and decision options when policy requires my approval, so that I can make a traceable decision. Given a request that meets an agreed approval condition, when it is submitted for review, then the assigned approver can approve or reject with a recorded decision; rejected or overdue requests follow agreed handling.	TBD	TBD / approval policy
Procurement	Order handoff	US-05	As a procurement coordinator, I want approved request details handed to the purchasing process, so that an order can be placed without rekeying agreed information. Given an approved request, when the handoff succeeds, then a purchasing reference is linked to the request; when it fails, then the failure is visible and recoverable by an owner.	TBD	TBD / purchasing interface
IT fulfillment	Receipt and asset handoff	US-06	As an IT fulfillment coordinator, I want receipt and asset-assignment details linked to the request, so that the device can be prepared and handed to the employee. Given the device is received, when required asset details are recorded, then assignment and handoff tasks can be completed and the requester is notified according to agreed process.	TBD	TBD / inventory, asset source
Exception owner	Exception handling	US-07	As an assigned exception owner, I want non-standard or uncertain requests routed with the reason and available context, so that they receive the right review without bypassing controls. Given a request outside the confirmed standard path, when validation identifies the exception, then the workflow records its reason, owner, and outcome.	TBD	TBD / exception policy

Reusable user story template

Field	Completeforeachstory
User type / role	[Who needs the capability?]
Epic / functional requirement	[Outcome area or capability]
Story ID	[US-##]
User story	As a [role], I want [capability], so that [benefit].
Acceptance criteria	Given [context], when [event/action], then [observable outcome]. Add alternate, error, permission, and exception cases.
Priority / release	[Owner decision; release or pilot stage]
DFD traceability	[Process ID, data flow, data store, or interface affected]
Dependencies / owner	[Policy, data source, integration, team, and accountable owner]

Definition of ready for review

- The user role and desired outcome are specific; the story describes a need rather than a proposed technical implementation.
- Acceptance criteria are testable, including expected exception and failure behavior where relevant.
- Policy conditions, data sources, system ownership, and interfaces are confirmed or explicitly marked as dependencies.
- Story is linked to the relevant DFD process and data flow; priority and release have an accountable decision owner.

Open design questions

Question	Owner / answer
What exactly qualifies as a standard laptop request, and who owns eligibility and approval rules?	[TBD]
Which systems are authoritative for worker data, catalog, inventory, purchase status, and asset assignment?	[TBD]
Which actions may proceed automatically, and which always require human review?	[TBD]
What data is necessary at each handoff, and what access, retention, and audit controls apply?	[TBD]
How will duplicate requests, integration outages, retries, and manual recovery be handled?	[TBD]
What baseline and success measures will be used for the pilot?	[TBD]